

Student Marks & Grading System

A simple Python-based console application to manage student marks and academic results.

Student Details

- **Name:** Keshav Chavra
- **Registration No.:** 26MIM10251
- **Course:** Python Essential
- **Program:** Integrated M.Tech – Artificial Intelligence
- **Year:** 1st Year
- **College:** VIT Bhopal University

1. Overview of the Project

The **Student Marks & Grading System** is a simple Python-based console application developed to manage student marks and academic results.

The program allows the user to enter a student's **name, roll number, and marks in five subjects** — Maths, English, Hindi, Physics, and Chemistry. It automatically calculates the **total marks and percentage**, assigns a **grade**, and determines the student's **PASS/FAIL status**.

In addition to result calculation, the system provides basic student record management operations such as **add, display, search, update, delete, and class summary**.

The project is designed as a beginner-friendly Python application and was developed and tested in **Google Colab**.

2. Features

- **Add Student:** Enter a student's name, roll number, and marks.
- **Input Validation:** Accept marks only from 0 to 100 and handle non-numeric input.
- **Automatic Calculation:** Calculate total marks and percentage.
- **Grade Assignment:** Assign a grade from A+ to F.
- **Pass/Fail Result:** A student passes only if they score at least 40 in every subject.
- **Display Records:** View all student records.
- **Search Student:** Find a student using their roll number.
- **Update Student:** Update a student's name and marks.
- **Delete Student:** Remove a student record.
- **Class Summary:** Show the number of students, pass/fail counts, class average, highest percentage, and top student's name.
- **Menu Validation:** Show a message when an invalid menu option is entered.

3. Technologies / Tools Used

- Python
- Google Colab
- GitHub
- Variables and data types
- Conditional statements
- for and while loops
- Functions
- Lists and dictionaries
- Input validation
- Basic calculations and data processing

4. Steps to Install and Run the Project

1. The program displays a menu of available operations.
2. The user selects an option.
3. To add a student, the program asks for the student's name and roll number.
4. The program checks whether the roll number already exists.
5. The user enters marks for all five subjects.
6. The program validates each mark.
7. It calculates the total and percentage.
8. It assigns a grade and checks the pass/fail condition.
9. The student record is added to the program's list.
10. The user can then display, search, update, or delete records, or view the class summary.
11. The menu repeats until the user chooses Exit.

5. Instructions for Testing

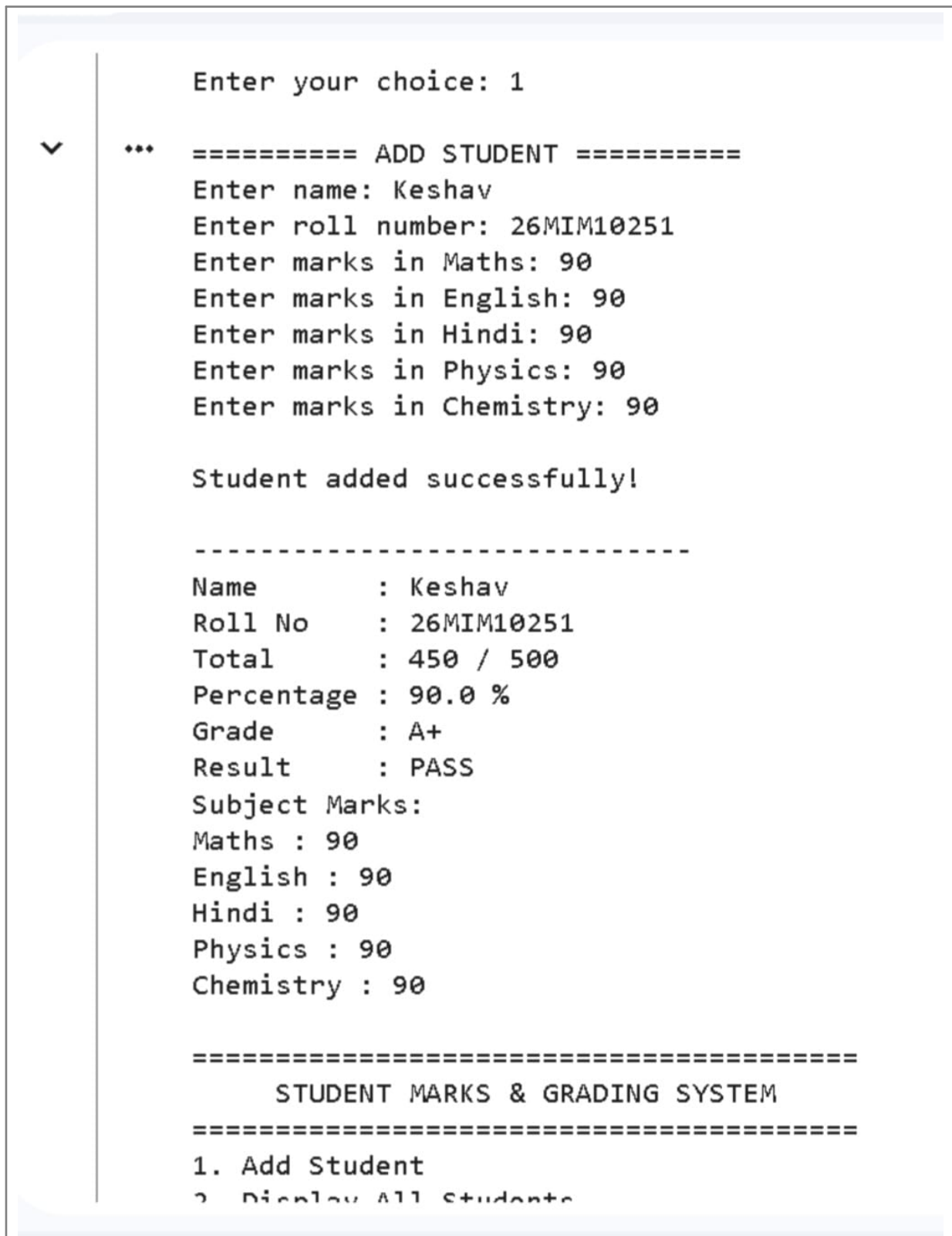
The program was checked using the following operations:

- Adding a student record
- Displaying all student records
- Searching for a student by roll number
- Updating a student record
- Deleting a student record
- Entering an invalid menu choice
- Checking marks validation
- Viewing the class summary
- Exiting the program

6. Screenshots

The following pages show the program output for each tested operation.

Screenshot 1: Add Student



Screenshot 2: Display All Students

```
=====
      STUDENT MARKS & GRADING SYSTEM
***  =====
      1. Add Student
      2. Display All Students
      3. Search Student
      4. Update Student
      5. Delete Student
      6. Class Summary
      7. Exit

Enter your choice: 2

===== ALL STUDENTS =====

-----
Name       : Keshav
Roll No    : 26MIM10251
Total      : 450 / 500
Percentage : 90.0 %
Grade      : A+
Result     : PASS
Subject Marks:
Maths      : 90
English    : 90
Hindi      : 90
Physics    : 90
Chemistry   : 90

=====
      STUDENT MARKS & GRADING SYSTEM
=====
```

Screenshot 3: Search Student

```
=====
      STUDENT MARKS & GRADING SYSTEM
=====
***
1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Summary
7. Exit

Enter your choice: 3

===== SEARCH STUDENT =====
Enter roll number: 26MIM10251

Student Found!

-----
Name       : Keshav
Roll No    : 26MIM10251
Total      : 450 / 500
Percentage : 90.0 %
Grade      : A+
Result     : PASS
Subject Marks:
Maths      : 90
English    : 90
Hindi      : 90
Physics    : 90
Chemistry  : 90
```

Screenshot 4: Update Student

```
Enter your choice: 4

***
===== UPDATE STUDENT =====
Enter roll number to update: 26MIM10251
Enter new name: Keshav Chavra
Enter marks in Maths: 90
Enter marks in English: 90
Enter marks in Hindi: 90
Enter marks in Physics: 90
Enter marks in Chemistry: 90

Student updated successfully!

-----
Name       : Keshav Chavra
Roll No    : 26MIM10251
Total      : 450 / 500
Percentage : 90.0 %
Grade      : A+
Result     : PASS
Subject Marks:
Maths      : 90
English    : 90
Hindi      : 90
Physics    : 90
Chemistry  : 90

=====
          STUDENT MARKS & GRADING SYSTEM
=====
1. Add Student
```

Screenshot 5: Invalid Menu Choice and Delete Student

```
=====
      STUDENT MARKS & GRADING SYSTEM
** =====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Summary
7. Exit

Enter your choice: 90

Invalid choice! Please select 1 to 7.

=====
      STUDENT MARKS & GRADING SYSTEM
=====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Summary
7. Exit

Enter your choice: 5

===== DELETE STUDENT =====
Enter roll number to delete: 26MIM10251
Student deleted successfully.
```

Screenshot 6: Delete Student and Class Summary

```
=====
      STUDENT MARKS & GRADING SYSTEM
***  =====
      1. Add Student
      2. Display All Students
      3. Search Student
      4. Update Student
      5. Delete Student
      6. Class Summary
      7. Exit

Enter your choice: 5

===== DELETE STUDENT =====
Enter roll number to delete: 26MIM10251
Student deleted successfully.

=====
      STUDENT MARKS & GRADING SYSTEM
=====
      1. Add Student
      2. Display All Students
      3. Search Student
      4. Update Student
      5. Delete Student
      6. Class Summary
      7. Exit

Enter your choice: 6

===== CLASS SUMMARY =====
No student records found
```

Screenshot 7: Class Summary and Exit

```
Enter your choice: 6
...
===== CLASS SUMMARY =====
No student records found.

=====
      STUDENT MARKS & GRADING SYSTEM
=====
1. Add Student
2. Display All Students
3. Search Student
4. Update Student
5. Delete Student
6. Class Summary
7. Exit

Enter your choice: 7

Thank you for using Student Marks & Grading System!
```
